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ASDP 2026 SINGAPORE

3RD ANNUAL CONGRESS OF
ASIAN SOCIETY OF DIGITAL PATHOLOGY

INTELLIGENCE IN EVERY PIXEL:
POWERING NEXT GENERATION PATHOLOGY

1 - 3 NOVEMBER 2026 | ACADEMIA, SINGAPORE GENERAL HOSPITAL

PathCoD

Pathology Chain-of-Discussion for Concept-Guided Multi-Agent Whole Slide Image Diagnosis

By

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Oral presentation · ASDP 2026, Singapore

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Why Is Whole-Slide Image Diagnosis Challenging?

A reliable system must analyse complementary cues, integrate evidence across regions and resolve ambiguity.

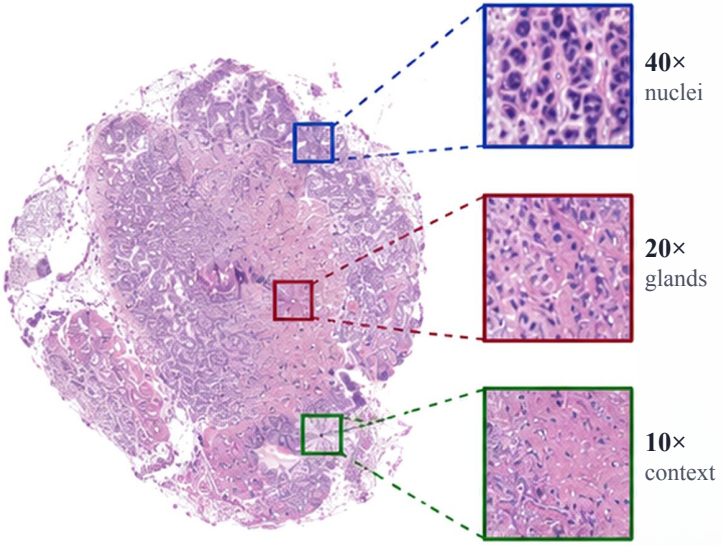


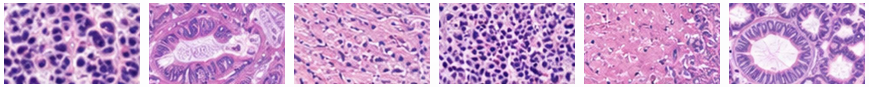
Figure 1: one H&E whole slide image; each diagnostic cue is visible only at its own magnification.

1. Gigapixel scale

~100,000 × 100,000 px per slide: tissue context lives at 10×, nuclear detail at 40×.

2. Multiple complementary cues

Diagnosis integrates patterns spread across different regions.



Nuclei | Glands | Stroma | Inflammation | Necrosis | Normal

3. Cues, and experts, disagree

Morphological cue	Grade it suggests
Nuclear atypia	Grade 3
Gland architecture	Grade 2
Stromal reaction	Grade 2

Three pathologists grade the same case 2, 3 and 2 (illustrative): inter-observer disagreement is common in borderline cases.

From One Generalist Model to Collaborating Specialist Agents

Same slide, same task (tumour grading): one model reading region by region vs. concept agents that debate the evidence.

(a) Generalist, sequential reasoning



Low mag.: locate a region

Medium mag.: zoom into it

High mag.: decide from it

Prediction: poorly differentiated (G3), incorrect

Anchored on the most atypical field; overall gland formation is never weighed.

- Misses complementary cues elsewhere on the slide
- No mechanism to resolve conflicting evidence
- One opaque decision, hard for a pathologist to audit

(b) PathCoD: concept-guided multi-agent discussion, six specialists, three consulted



Six concept agents, each reading the whole slide

1. A1 (nuclear): “Marked pleomorphism and prominent nucleoli in a few foci. Favours high grade.”
2. A3 (glands): “Disagree: most of the tumour still forms glands, so the overall grade is lower.”
3. A5 (stroma): “Irregular glands in desmoplastic stroma confirm invasion; supports A3.”

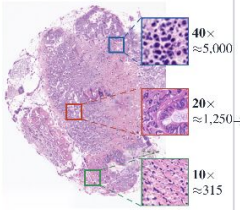
Consensus: moderately differentiated (G2), correct

Grade set by gland formation; focal high-grade atypia flagged; every claim traceable to a region.

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Stage 1 · Multi-teacher knowledge extraction and fusion (offline, cached once)

Multi-scale WSI $\mathscr{W}$



224×224 patches · tissue mask · stain norm.

Frozen pathology FMs T_m

- WSI-LLaVA — ViT-L + LLaMA-7B
- CONCH — ViT-B
- Quilt-LLaVA — multimodal
- H-Optimus-0 — ViT-Giant

$N_\ell \times d_m$ → PCA → $N_\ell \times 768$ (run at each scale $\ell = 1,2,3$; >95% variance)

ℓ_2-norm ↓ mean over $M=4$ teachers

$e^{(i,\ell)} \in \mathbb{R}^{768}$ → 40× / 20× / 10× → concat → $\mathbb{R}^{2304}$ → Linear 2304→768 → $F_{\text{fused}}, N \times 768$ → cached

Cached teacher signals (per patch, per scale)

- Patch embedding $e_m^{(i,\ell)} \in \mathbb{R}^{d_m}$ → fusion, NMF
- Soft prediction $\pi_m^{(i)}(\tau{=}3)$ → $\mathscr{L}_{\text{resp}}$
- CLS-to-patch attention $\alpha_m^{(i)} \in \mathbb{R}^{196}$ → $\mathscr{L}_{\text{attn}}$
- Mid-layer feature layer $\lfloor L_m/2 \rfloor$ → $\mathscr{L}_{\text{feat}}$

Frozen · Trained (ours) · Fit once

$X_{\text{train}} \in \mathbb{R}^{N \times 768}$, $N \approx 50$M patches

Stage 2 · Concept discovery (NMF)

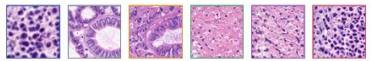
$X \approx W \times H$ — X: $N \times 768$; W: $N \times K$; H: $K \times 768$ rows h_k

$K{=}6$ · RE 0.18 · Sil. 0.62 · pathol. >7.5

$$\min_{W,H \geq 0} \|X - WH\|_F^2 + \lambda_W \|W\|_1 + \lambda_H \|HH^\top - I\|_F^2$$

sparse W ($\lambda_W{=}0.01$), orthogonal H ($\lambda_H{=}0.1$)

Discovered concepts



$\kappa = 0.78$ vs. pathologists

Stage 3 · Specialist agents (ViT-S, 22M)

Concept-guided sampling — $p_k(i) \propto w_{ik}^2$

Six specialist agents A_k

- A_1 Nuclear morphology — ViT-S
- A_2 Normal tissue — ViT-S
- A_3 Tissue architecture — ViT-S
- A_4 Necrosis — ViT-S
- A_5 Stromal microenv. — ViT-S
- A_6 Inflamm. infiltrate — ViT-S

Inside agent A_k

3 scales → ViT-S per scale → concept attention query $= h_k$ → MLP fuse $z_k \in \mathbb{R}^{384}$ → softmax head π_k → $\hat{y}_k$, c_k

single agent: 82.3–85.7% · disagree on 37% of cases

Stage 4 · Query-aware routing and Chain-of-Discussion (CoD)

Query-aware router

Query q "What is the tumour grade?" → BioBERT; 512^2 → EfficientNet-B0 → Cross-attention + residual → MLP → s_k → Gumbel top-3 of 6 ($\tau \to 0.5$)

RL reward $R = \mathbb{I}[\hat{y}{=}y] - 0.1\,|\mathscr{A}_q|/K$

Independent assessment (example)

- Nucl. (A_1) G3, c=0.72
- Arch. (A_3) G2, c=0.68
- Strom. (A_5) G2, c=0.41

disagree? $\phi > 0.1$ — no / yes

Chain-of-Discussion round $t \leq 5$

1 message · 2 attend · 3 re-fuse · 4 update

$S^{(t-1)}$ → cls → MLP → $\gamma{=}0.7$; $\alpha_{k,k'}$; own $z_k^{(0)}$ (anchor); no: $t \leftarrow t{+}1$

stop if $\phi < 0.1$, $t = 5$, or stagnation → stop? — yes

41.3% skip CoD

Confidence-weighted vote

$w \propto c_k$ → Σ → $\hat{y}$, c_{final}

abstain if $c_{\text{final}} < 0.65$

Stage 5 · Training objectives

Agent distillation λ_{d}

Response KL (τ=3)	0.40
Hard labels	0.30
Feature matching	0.20
Attention transfer	0.08
Concept consistency	0.02

CoD: $\mathscr{L}_{\text{ace}} + 0.1\,\mathscr{L}_{\text{eff}} + 0.05\,\mathscr{L}_{\text{agree}} + 0.1\,\mathscr{L}_{\text{cal}} + 0.02\,\mathscr{L}_{\text{comm}}$

Schedule: cache → distil → RL routing → CoD → joint fine-tuning

Output

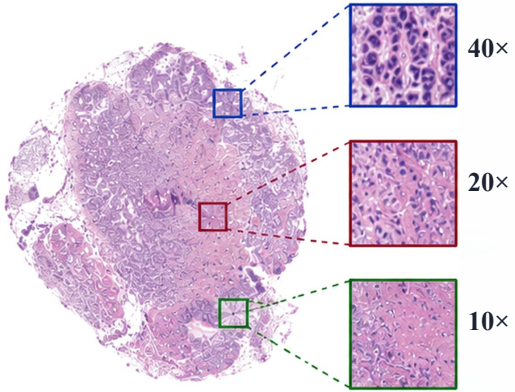
Grade 2 $c = 0.892$

$\hat{y}$ diagnosis · c confidence · e discussion trace · F abstain flag

Inside an Agent: From WSI to Evidence

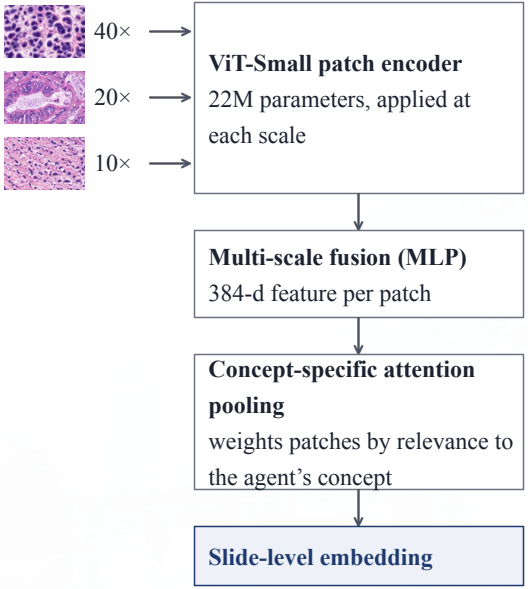
Each selected agent reads the slide at three scales, attends to its own concept, and returns a grade, a confidence and an evidence map.

(a) Multi-resolution input



224 × 224 patches at 40× (nuclei), 20× (architecture) and 10× (stromal context).

(b) Agent encoder



(c) Outputs for the grading query

Agent	Prediction	Conf.	Evidence
A1 Nuclear atypia, pleomorphism	Grade 3	0.72	
A3 Architecture gland formation	Grade 2	0.68	
A5 Stroma desmoplastic reaction	Grade 2	0.41	

Alone, agents reach 82.3–85.7% and disagree on 37% of cases. Here: Grade 3 vs Grade 2, so discussion is triggered.

A Worked Example: Grade 3 or Grade 2?

An illustrative paraphrase of the latent exchange; the real messages are 256-d vectors.

Agent (start)	Round 1: share evidence	Round 2: challenge	Round 3: converge
A1 Nuclear (G3, 0.72)	Enlarged, pleomorphic nuclei. I say Grade 3.	Agreed, atypia is moderate overall.	Updating to Grade 2; focal atypia flagged.
A3 Architecture (G2, 0.68)	Most of the tumour forms glands. Grade 2.	A1, the high-grade nuclei are only focal.	Grade 2, confidence increased.
A5 Stroma (G2, 0.41)	Desmoplasia confirms invasion; grade-neutral.	Stroma does not set grade; I defer to A3.	Grade 2. Converged.

G2

Moderately differentiated · confidence 89.2%

A1 revised Grade 3 → Grade 2. Gland formation decides; focal high-grade atypia is flagged for the pathologist.

Like a tumour board: conflicts surface, complementary evidence is weighed, and the reasoning can be audited.

Full Comparison on TCGA

Best on every accuracy and calibration metric, at a fraction of the cost of larger models.

Table 1. TCGA test set (1,626 WSIs, 32 subtypes). Mean ± std over five seeds; time (s), memory (GB) and TFLOPs per WSI.

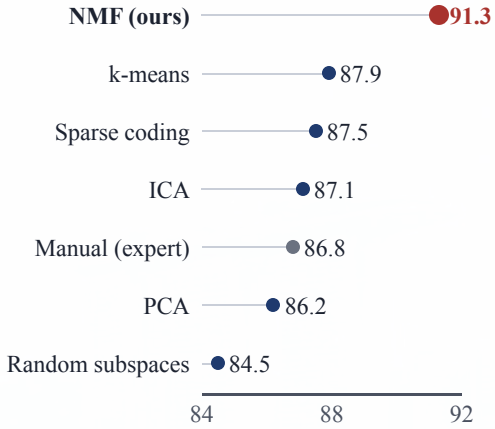
Type	Method	Params	Acc ↑	BAcc ↑	F1 ↑	AUC ↑	κ ↑	ECE ↓	Time ↓	Mem. ↓	TFLOPs ↓
Vision FM	CONCH	307M	84.9 ± 0.5	83.2	0.842	0.958	0.843	0.081	32.7	14.8	5.1
	UNI	307M	86.2 ± 0.4	84.7	0.856	0.963	0.856	0.075	34.5	16.1	5.8
	Virchow2	632M	87.9 ± 0.4	86.5	0.873	0.970	0.873	0.069	39.6	21.3	7.4
	GigaPath	1.1B	88.1 ± 0.5	86.8	0.875	0.972	0.875	0.071	45.2	26.7	9.1
	H-Optimus-0	1.1B	88.7 ± 0.4	87.4	0.881	0.974	0.881	0.067	47.8	28.4	9.6
VLM	WSI-LLaVA	7.0B	86.8 ± 0.3	85.3	0.862	0.966	0.862	0.078	60.4	37.9	14.2
	PathChat	13B	88.2 ± 0.5	86.9	0.876	0.972	0.876	0.066	58.3	33.4	13.1
Compact	H0-mini	86M	85.7 ± 0.7	84.1	0.851	0.960	0.851	0.074	24.7	10.9	3.8
Multi-agent	WSI-Agents	28B	89.0 ± 0.4	87.7	0.885	0.976	0.885	0.061	72.6	41.2	18.5
Ours	Teacher ensemble (4 FMs)	8.7B	90.1 ± 0.3	88.9	0.896	0.979	0.896	0.051	51.9	29.8	10.8
	PathCoD w/o CoD	259M	86.1 ± 0.5	84.6	0.855	0.961	0.855	0.089	12.4	8.9	2.6
	PathCoD (full)	**259M**	**91.3 ± 0.3*** **	**90.1**	**0.909**	**0.982**	**0.905**	**0.045**	**19.7**	**10.6**	**3.4**

*** $p < 0.001$ vs. H-Optimus-0 (paired t-test, Bonferroni-corrected). 259M parameters cover all six specialists; 190M are active per case.*

Ablations: Concepts, Teachers, Routing

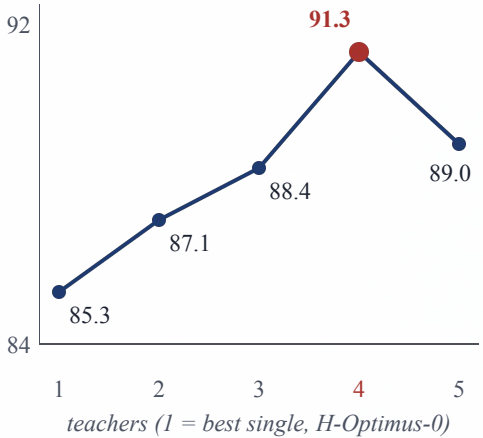
Discovered concepts beat expert-defined roles, teacher diversity beats teacher count, and three agents beat all six.

(a) How concepts are found



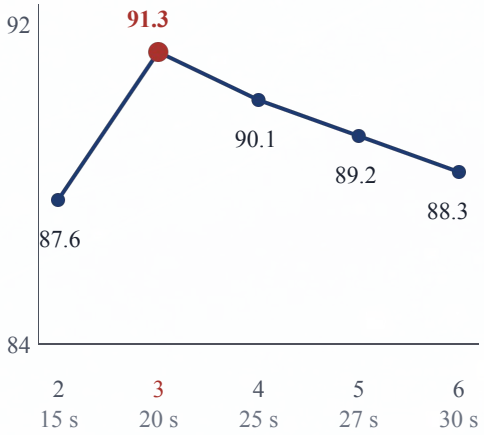
+4.5 over expert-defined roles

(b) Number of teachers



Adding GigaPath as a fifth **hurts** (89.0%)

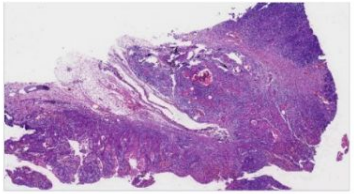
(c) Agents activated



Three agents: best accuracy, a third less time than six

Takeaway

Input



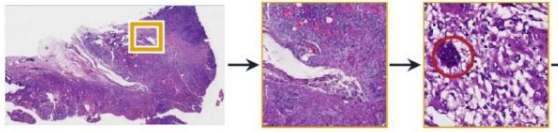
whole slide image (WSI)

Question: *What is the most likely diagnosis for this WSI?*

Ground truth:
Squamous cell carcinoma

(a) Conventional: one model, one sequential zoom path

1 locate (low mag.) → 2 zoom in → 3 inspect (high) → 4 reason and commit



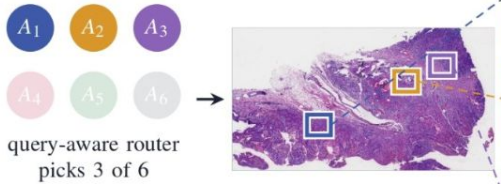
Sees gland-like spaces in this one field, and concludes adenocarcinoma.

→ **Adenocarcinoma** ✗ incorrect

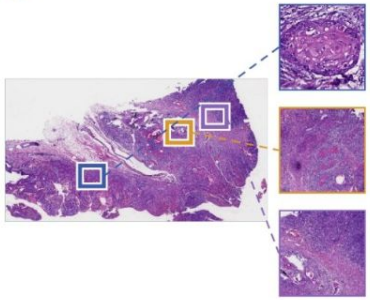
Why it fails
- one path, fixed early
- rest of slide unseen
- misread cue unchallenged

(b) PathCoD (ours): concept specialists that read in parallel and discuss

1 select → 2 read the slide in parallel → 3 Chain-of-Discussion → 4 decide



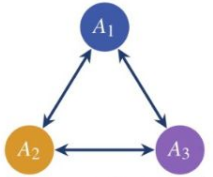
query-aware router picks 3 of 6 concept specialists



A_1 **Nuclear**
large atypical nuclei, dense eosinophilic cytoplasm

A_2 **Architecture**
solid nests, keratin pearls

A_3 **Stromal**
desmoplastic stroma, gland-like spaces only focal



exchange, challenge, refine (latent messages, ≤5 rounds): *gland-like focus is judged focal, not representative*

→ **Squamous cell carcinoma** ✓ correct

with confidence, a discussion trace, and the option to abstain

Thank You!

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Scan to connect